

PRAVEEN ANTONY

PRINCIPAL ENGINEER | AUTONOMOUS SYSTEMS ARCHITECT (AGENTIC AI & ORCHESTRATION)

Location: Apex, NC | Phone: 469-231-8419 | [LinkedIn](#) | [Portfolio](#) | [GitHub](#)

PROFESSIONAL SUMMARY

Principal Engineer with 17+ years of experience architecting enterprise-scale cloud-native platforms, distributed systems, and modernization programs across regulated financial services environments. Lead technical strategy for business-critical platforms, with deep experience in Java, Spring, API-first architecture, event-driven systems, cloud modernization, and engineering leadership. Over the last few years, expanded this foundation into Agentic AI and enterprise AI engineering using LLMs, Spring AI, RAG, and MCP to deliver workflow automation and intelligent platform capabilities.

TECHNICAL SKILLS

AI & AGENTIC ENGINEERING

Large Language Models (LLMs) • Agentic AI • Multi-Agent Systems • LangGraph • Model Context Protocol (MCP) • MCP Clients/Servers • MCP Host Integrations • Tool Calling • RAG • FAISS (Facebook AI Similarity Search) • Vector Databases • Spring AI • LangChain • LangSmith • OpenAI Agents SDK • CrewAI • AutoGen • Prompt Engineering • Context Engineering • AI Evaluation • Persistent Memory Architecture • Token Management • Generative AI • AI Agents • Python • FastAPI • OpenTelemetry

CLOUD & PLATFORM ARCHITECTURE

AWS • Kubernetes (EKS) • Java 21 • Spring Boot • Microservices • API-First Architecture • Event-Driven Architecture • REST APIs • GraphQL • Nx Monorepo Architecture • Distributed Systems • AWS Lambda • Terraform • Helm • Docker • Docker Compose • PostgreSQL • Node.js • CI/CD • Jenkins • GitLab

FRONTEND & SECURITY

Angular • TypeScript • React • RxJS • NgRx • Module Federation • React Native • JavaScript • Tailwind CSS • OAuth2 • OpenID Connect (OIDC) • JWT • Datadog • Splunk • Sentry • Encryption

PROFESSIONAL EXPERIENCE

FIDELITY INVESTMENTS

Principal Software Engineer / Technical Lead

June 2023 – Present | Raleigh, NC

- Lead architecture and delivery for enterprise financial platforms, partnering with architects and product leaders on multi-year modernization strategy.
- Design cloud-native architectures using Java, Spring Boot, Angular, AWS, Kubernetes, GraphQL, and event-driven microservices.

- ♦ Drive migration of legacy applications into secure, scalable, **API-first platforms** with enterprise standards for security, observability, governance, and CI/CD.
 - ♦ Lead cross-team platform initiatives that improve developer productivity, architecture consistency, and reusable engineering capabilities across the organization.
 - ♦ Contributed as a monorepo and GraphQL champion, helping drive reusable standards, governance, and developer enablement across teams.
 - ♦ Over the last few years, led enterprise **Agentic AI** initiatives enabling intelligent workflows and AI-powered business applications.
 - ♦ Architected an **Agentic Experience Workspace & Orchestration Layer**, building an Angular Agent Workspace and a Spring AI-based Life Events Agent over existing DEC and domain API ecosystems.
 - ♦ Designed reusable enterprise architecture patterns for **LLMs, RAG, AI agents, memory, and orchestration frameworks**.
 - ♦ Lead development of internal AI accelerator platforms including **DP Labs Platform, LeapDev AI Platform, ConfluxAI, and FlowPilot**.
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EBSCO INFORMATION SERVICES

Senior Software Engineer II

March 2017 – June 2023

- ♦ Designed and delivered scalable enterprise platforms supporting global products and high-volume users.
 - ♦ Led application modernization initiatives, architecture improvements, and engineering best practices.
 - ♦ Built full-stack cloud-native solutions using Java, Angular, Node.js, REST APIs, and automated CI/CD pipelines.
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RANDSTAD TECHNOLOGIES

Senior Software Developer

September 2014 – March 2017

Clients: Fidelity Investments, Ally Financial

- ♦ Served on enterprise Technical Review Board evaluating solution architecture and engineering standards.
 - ♦ Developed scalable financial applications using **Java, Node.js, Angular, and REST services**.
 - ♦ Delivered reusable frontend architectures for high-traffic enterprise systems.
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KFORCE INC.

Senior UI Developer

October 2013 – September 2014

Client: Sparta Systems

- ♦ Developed enterprise web applications and reusable UI component frameworks.
 - ♦ Contributed to frontend modernization and application architecture improvements.
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INFOSYS LTD.

Technology Analyst

September 2008 – October 2013

- ♦ Developed enterprise integration solutions using Java, REST services, and scalable application frameworks.
- ♦ Delivered technology solutions for global enterprise clients.

ENTERPRISE PLATFORM & AI INITIATIVES

DP Labs Platform

Shared GenAI developer platform unifying AI services, connectors, governance controls, and modular UI capabilities for enterprise experimentation and reusable delivery.

Agentic Experience Workspace & Orchestration Layer

Built an Angular Agent Workspace and Spring AI-based orchestration agent that makes existing modeling workflows conversational by leveraging established DEC and domain APIs.

LeapDev AI Platform

Enterprise AI software engineering platform enabling AI-assisted planning, implementation, code review, CI/CD automation, and pull request generation.

ConfluxAI

Enterprise **RAG-powered knowledge platform** enabling conversational search across engineering documentation.

FlowPilot

AI-driven incident response platform integrating ServiceNow, Jira, Microsoft Teams, and human approval workflows using **LangGraph** and **MCP**.

PERSONAL AI PROJECTS

Dev Agent & PR Generator

Multi-agent software engineering system generating production-ready pull requests using Planner, Developer, Reviewer, and CI/CD agents.

DealScoutAI

LLM-powered intelligence platform analyzing CRM, email, and calendar data.

Soccer Bot Team Selector

Intelligent automation system for team selection workflows with backend scheduling, skill-based balancing algorithms, and resilient state management to overcome cold-start and ephemeral-storage constraints.

Bedtime Story Architect

Multimodal AI application generating personalized narrated and illustrated stories.

Fifty Six Card Game

AI-powered Progressive Web App featuring intelligent gameplay, probability-based decisions, and multiplayer support.

EDUCATION & CERTIFICATIONS

Bachelor of Technology (B.Tech.)

Computer Science & Engineering

Certifications

- ♦ AWS Certified Solutions Architect – Associate
- ♦ Apollo Graph Developer – Professional
- ♦ Scaled Agile Framework (SAFe)